

Question:

Two walls and the ceiling of a room meet at right angles at point \$P\$. A fly is in the air one meter from one wall, eight meters from the other wall, and nine meters from point \$P\$. How many meters is the fly from the ceiling?

(A) $\sqrt{13}$ (B) $\sqrt{14}$ (C) $\sqrt{15}$ (D) 4
(E) $\sqrt{17}$

Reasoning + Answer:

<think>

1. Understanding the Problem :

- We have a room where two walls and the ceiling meet at a right angle at point (P) .
- The fly is in the air , one meter from one wall , eight meters from the other wall , and nine meters from point (P) .
- We need to find the distance from the fly to the ceiling .

2. Setting Up the Coordinates :

- Let the walls be (x) - and (y) - axes , and the ceiling be the (z) -axis .
- Let the fly be at point $(A(x, y, z))$.
- Given :
 - $(d(x, 0) = 1)$ (distance from wall at $(x = 0)$ to the fly)
 - $(d(0, y) = 8)$ (distance from wall at $(y = 0)$ to the fly)
 - $(d(0, 0, z) = 9)$ (distance from point (P) to the fly)

3. Using the Distance Formula :

- The distance formula in 3D is $(d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2})$.

4. Formulating the Equations :

- For the distance from wall at $(x = 0)$:

$$\sqrt{x^2 + y^2 + z^2} = 9$$

- For the distance from wall at $(y = 0)$:

$$\sqrt{x^2 + z^2 + y^2} = 9$$

5. Simplifying the Equations :

- From the first equation :

$$x^2 + y^2 + z^2 = 81$$

- From the second equation :

$$x^2 + y^2 + z^2 = 81$$

6. Using the Given Coordinates :

- We know $(x = 1)$:

$$1^2 + y^2 + z^2 = 81 \implies 1 + y^2 + z^2 = 81 \implies y^2 + z^2 = 80$$

7. Finding the Distance to the Ceiling :

- The distance to the ceiling is (z) .
- Using the given $(z = 9)$:

$$z^2 = 9^2 = 81$$

8. Conclusion :

- The distance from the fly to the ceiling is $(z = 9)$, which is $(\sqrt{81} = 9)$.

Therefore , the answer is $(\boxed{E})$.

<answer> $(\sqrt{15})$ </answer>